

∴ By defn, each of the n columns of A are pivot columns.

So, T is one-to-one if and only if A has n pivot columns. $\square$

32. Let $T: \mathbb{R}^n \rightarrow \mathbb{R}^m$ be a linear transformation, with A its standard matrix. Complete the following statement to make it true: " T maps $\mathbb{R}^n$ onto $\mathbb{R}^m$ if and only if A has _____ pivot columns." Find some theorems that explain why the statement is true.

Solution: A has m rows and n columns from Theorem 10.

- By Theorem 12(a), T maps $\mathbb{R}^n$ onto $\mathbb{R}^m$ if and only if ~~the~~ the columns of A span $\mathbb{R}^m$.
- By Theorem 4, The columns of A span $\mathbb{R}^m$ if and only if each row of A has pivots.
- A can be converted to a ~~row~~ echelon form by the Row Reduction algorithm.
- By Condition 2 of echelon form, it's easy to see why there must be exactly m pivot columns.
- T maps $\mathbb{R}^n$ onto $\mathbb{R}^m$ if and only if A has m pivot columns. $\square$

35. If a linear transformation $T: \mathbb{R}^n \rightarrow \mathbb{R}^m$ maps $\mathbb{R}^n$ onto $\mathbb{R}^m$, can you give a relation between m and n ? If T is one-to-one, what can you say about m and n ?

Solution: From Problem 31, we know that if T is one-to-one, The standard matrix A has n pivot columns. From the defn. of the echelon form of the matrix, by Condition 2 and 3, we know that each of n columns has exactly 1 pivot. By defn, each row has exactly 1 pivot. $\square \therefore m \geq n$

From Problem 32, we know that if T maps $\mathbb{R}^n$ onto $\mathbb{R}^m$, then A must have m pivot columns. We know that each row has exactly 1 pivot, by defn. $\square \therefore m \leq n$.